

Decision theory introduced

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1. Kinds of decision theory

‘Decision theory’ covers debates across disparate domains, united by a similarity of formalism and terminology.

(i) Ethics. *Normative* decision theory is concerned with questions about what certain agents *ought* to do (in one or more senses of ‘ought’); or, what *mental states* they ought to be have (e.g. of preference and credence); or about what it takes for them to be *rational*.

(ii) Psychology/biology. *Descriptive* decision theory seeks accurate generalizations about what decisions people (and other animals) in fact make when they have multiple options available to them. Psychology and evolutionary biology may seek to understand the causal basis for, and evolutionary emergence of, the dispositions underlying such generalizations.

(iii) Social science. Often, a simple model that is not all that accurate at predicting individual behavior will still provide valuable understanding of large-scale patterns (where for whatever reason the details of the individual failures of the model are irrelevant). Economists and other social scientists (as well as behavioral ecologists) employ decision theory in this role.

(iv) Philosophy of mind. The classic tradition of decision theory going back to Ramsey and Savage suggests that the theory can shed light on questions in the philosophy of mind, such as what it is for an agent to have a certain *credence* that something is the case, or what it is for an agent to have a certain strength of *preference* for one thing’s being the case over another thing’s being the case. Although the general philosophical arguments against behaviorism arguably refute more straightforward ideas about how this works, the tradition of functionalism in the philosophy of mind suggests ways in which some insights might be salvaged from this program.

2. Normative expected-utility theory: the basic schema

There is an “orthodox” or “received” view in decision theory, namely *expected utility theory*. Its roots go back to Fermat, Pascal, and Bernouilli; these days it is often presented in a formalism pioneered by Savage (1954). One represents a specific decision situation as a matrix in which the columns are “states”, assigned nonzero “probabilities” that add up to 1; the rows correspond to “available acts” in the given situation; and the cells are *utilities*.

Here’s an example.

	Ball will land red 18/37	Ball will land black 18/37	Ball will land green 1/37	EU
Bet all on red	100	-100	-100	-100/37
Bet all on black	-100	100	-100	-100/37
Don't bet	0	0	0	0

The values in the “EU” column are calculated by taking a weighted average of the utilities in the cells in the given row, using the probabilities of the states as the weights:

$$EU(a) = \sum_s u(a, s)P(s)$$

The signature claim of EU theory, when understood as a normative theory of ‘how one ought to act’, is that the agent *ought not* to perform any of the actions whose EU value is lower than that of some other action.

- Some also understand “EU theory” to include the much less important claim that none of the actions with maximal EU is such that one ought not to do it. Assuming that it never happens that every available act is such that one ought not to perform it, this further claim is only relevant in cases where there are ties.

To ask whether normative EU theory is *true* we’ll need to settle a bunch of hard interpretative questions.

- What are the “states” (for a given decision-situation)?
- What are the “probabilities” assigned to the states?
- What are the “acts”?
- What determines the “utility” assigned to a given combination of state and act?
- In what sense of ‘ought’ or ‘rational’ is it being claimed that the actions with non-maximal EU ought not to be performed, or are not performed by rational agents?

3. Dominated options

In wrestling with these questions, it is worth noting that there are some cases where one does not need to consult the probabilities at all to see that EU theory will deliver the result ‘this act ought not to be done’. Act *a* dominates act *b* iff $u(a, s) \geq u(b, s)$ for every state *s* and $u(a, s) > u(b, s)$ for some state *s*. Assuming every state receives nonzero probability, this is sufficient for it to be the case that $EU(a) > EU(b)$, and thus for normative EU theory to yield the verdict ‘*a* ought not to be done’.

	Raining	Not raining
Take subway	2	2
Walk	1	3
Skip class	0	0

A relatively weak claim entailed by EU theory (if we build in ‘all states have nonzero probability’): any action that is dominated ought not to be performed.

A strong claim inconsistent with EU theory (and presumably obviously false on any sense of ‘ought’ for which EU theory isn’t obviously false): no action that isn’t dominated ought not to be performed.

Note that all that matters to whether an option is dominated is the *ordering* of the utility values. So, finding an interpretation of all this jargon that makes ‘dominated options ought not to be performed’ true is a less challenging task than finding an interpretation that makes EU theory true.

4. What are the “utilities” and what is the “ought”?

Decision theory was originally developed as a guide to gamblers, with utilities identified with monetary gains and losses. But this was surely never intended as a general theory—there is surely no interesting sense of ‘ought’ in which you ought to cut off your arm if you’re convinced you can get a dollar by doing so.

A somewhat less silly interpretation identifies the utility scores with the agent’s total net life happiness—something like the net balance of pleasure over pain over the course of their life. But this theory is also pretty terrible. Some people care deeply about others, and knowingly sacrifice their own happiness, even their lives, to help those they care about. I doubt there is any interesting sense of ‘ought’ in which these people ought not to sacrifice themselves like that, or any interesting sense of ‘irrational’ in which it is *irrational* for them to sacrifice themselves.

- Sometimes, people try to save something along these lines by saying that the theory is only concerned with what one ought to do in special choice situations where one can have no effect on the well-being of any other agents. But such choice situations would seem to be vanishingly rare, and it’s unclear why we’d care about a theory restricted to them.

An interpretation that escapes this objection is ‘EU utilitarianism’, where the utility value of an act, state pair is identified with something like the total net happiness of all agents in the universe. This is defensible, though obviously highly controversial, if one also interprets the ‘ought’ as something ethical. It’s less defensible as a theory of “ideal rationality”: it sounds pretty tendentious to say that selfish people who pursue their

own happiness over those of others are being *irrational*. On the way of thinking of “rationality” dominant in decision theory, ideally rational people could pursue totally ridiculous goals, so long as they choose the means that (according to their credences) are best suited to advancing those goals: “’tis not contrary to reason, to prefer the destruction of the world to the scratching of my little finger” (Hume).

The most popular interpretation of the formalism fits with this, by interpreting the utility values as a measure of the strength of the agent’s *preferences* among different outcomes/possible worlds, at the time of decision. (So, Hume’s agent would have a utility function that assigns higher numbers to worlds where the world is destroyed than to worlds where their little finger is scratched.)

- If there is a sense of ‘ought’ that fits with the preference-theoretic interpretation of EU theory it is obviously not the sort of thing typically in play in ethics, nor does it have much to do with the traditional virtue of ‘prudence’. Work on the semantics of ‘ought’ suggests it’s may well be flexible enough to include a sense in which Hume’s agent ought to destroy the world given a choice between that and the scratching of their finger.

There’s a related interpretation of EU theory as of (ideal) rationality which doesn’t use any word like ‘prefer’. It says that necessarily if you are ideally rational, there is *some* random variable u (a function from possible worlds to real numbers?) such that you are disposed, faced with any decision problem, to choose one of the actions that maximize the expected value of u (according to your credences?).

- There may not be a huge gulf between this and a preference-theoretic interpretation. According to a kind of behaviorism that’s been influential in the history of decision theory, there is nothing more to a random variable’s “representing your preferences” than your being disposed to maximize its expectation of it, so it’s *impossible* to be disposed to maximize a random variable that isn’t a measure of the strength of your preferences. And even functionalists might agree that things would have to be going wrong in some very strange and specific way for an agent to be stably disposed to act so as to maximize the expectation of a random variable that’s drastically out of whack with their preferences.

Suppose we *are* interested in the kind of interpretation of ‘ought’ used in ethics, but are persuaded by the usual objections to utilitarianism. Does that mean that we have to give up on interpreting EU theory as a theory of the ethical ‘ought’? No: we might still accept the weaker existential claim (‘EU consequentialism’) that there exists a random variable (function from worlds to real numbers) such that necessarily, any act that does not maximise the expectation of that random variable (according to the agent’s credences) ought not to be performed. Or the even weaker existential claim (‘EU centred consequentialism’) that there exists a function from agents (and times?) to random vari-

ables such that necessarily, any act that does not maximize the expectation of the random variable that is the value of this function for the agent (and the time of decision) ought not to be performed. The latter claim is certainly consistent with all sorts of classic anti-utilitarian judgments (and the former claim may be too). Indeed, the worry that these claims are trivially true is arguably more pressing than the worry that they are false.

5. A note on ‘small world’ versus ‘big world’ decision problems

On pretty much any philosophically viable interpretation of ‘utility’, the utility value assigned to a state-of-affairs will have to be sensitive in a quite fine-grained way to the details of what’s going on there—even if the agent is monomaniacally fixated on having as high a bank balance as possible at some particular chosen time, the value of this will often depend on all sorts of factors beyond those represented in the simple decision matrices that get used in examples.

But one nice feature of EU theory (which distinguishes it from some competitors) is that it lets us ignore a whole lot of our uncertainty about utility so long as we don’t take it to be correlated with our current action.

For example: suppose that in addition to being uncertain where the roulette ball will land, you’re also uncertain whether (D) your rich uncle will die and bequeath you \$1000. So your *real* decision matrix is this:

	Red $\wedge$ D	Black $\wedge$ D	Green $\wedge$ D	Red $\wedge\neg$ D	Black $\wedge\neg$ D	Green $\wedge\neg$ D
Bet all on red	\$1,100	\$900	\$900	\$100	-\$100	-\$100
Bet all on black	\$900	\$1,100	\$900	-\$100	\$100	-\$100
Don’t bet	\$1,000	\$1,000	\$1,000	\$0	\$0	\$0

For each act A, we can break up $E_{Pr}(U(A))$ as $E_{Pr(\cdot | D)}(U(A))Pr(D) + E_{Pr(\cdot | \neg D)}(U(A))Pr(\neg D)$. But for each act A, $Exp_{Pr(\cdot | \neg D)}(U(A))$ is the same as $Exp_{Pr}(U(A))$ in the old, simpler decision problem, while $Exp_{Pr(\cdot | D)}(U(A))$ is this number plus \$1000. So the *real* EU of each act is just its EU in the simple problem + \$1000 $Pr(D)$: it’s thus harmless to use the simple representation when you’re deciding how to bet.

- So long as you value dollars linearly, this goes through even if you treat your uncle’s death as more likely given red than given black!

Beware that alternatives like ‘risk-weighted expected utility theory’ disrupt this simplification.

6. What is “probability”

The dominant view identifies the “probabilities” in the decision matrix with the *credences* of the agent at the time in question.

- The ‘ought’-theoretic interpretation struggles with cases where the agent’s credences don’t fit the axioms of probability (so that, e.g., their credences in the states sum to more than 1). One might try to get around this by instead identifying the probabilities with the credences the agent “ought” to have. But it’s controversial that there’s *unique* credence one ought to have.
- Some instead take themselves to grasp a notion of ‘evidential probability’ not understood in terms of credence, and use this in stating a decision theory.
- The ‘ideal rationality’ interpretation doesn’t face exactly this worry, since its proponent can say that any ideally rational agent does have credences that fit the probability axioms (and are defined on all relevant propositions.) It does however face worries about how such a demanding notion of ideal rationality is relevant to the evaluation of limited creatures like us.

There is also a rationality-theoretic interpretation of EU theory that doesn’t use the word ‘credence’: it says that necessarily if you are ideally rational, there is *some* probability function Pr such that you are disposed, faced with any decision problem, to choose one of the actions that maximize utility according to Pr .

- There may not be a huge gulf between this and a credence-theoretic interpretation. According to some behaviorists, there is nothing more to having a certain credence function than being disposed to maximize EU according to it, so it’s impossible to be disposed to maximize EU according to a probability function distinct from your credence function. And even functionalists might agree that things would have to be going wrong in some very strange and specific way for an agent to be stably disposed to maximize EU according to a probability function drastically out of whack with their credences.

7. What are the “states”?

Philosophers (like me) who are comfortable with ‘proposition’-talk can understand the “states” for a given decision problem as *propositions*. We might require them to always be chosen so as to be pairwise incompatible and jointly necessary; or require that the proposition that that more than one of them is true or that none of them is true have zero probability (in whatever we take to be the operative sense of ‘probability’); or at least require that the possibility of this not being the case be somehow negligible.

To get non-absurd results, it is crucial that the “states” should be propositions whose truth or falsity is (in some sense that remains to be clarified) *entirely beyond the agent’s*

control, at the time when the decision is being made. The following example due to Joyce (1999) illustrates what goes wrong if we don't impose this:

	Windshield broken	Windshield not broken
Pay protection	-\$510	-\$10
Refuse to pay	-\$500	\$0

There are two dominant philosophical approaches to cashing out this notion of 'entirely beyond the agent's control'. The tradition of "causal decision theory" (Stalnaker, Gibbard, Harper, Lewis, Skyrms, Eells...) brings in concepts like *causation*, *counterfactuals*, and/or *chance*. By contrast, according to "evidential decision theory", it's sufficient that the states be *probabilistically independent* of the acts (according to the agent's credences).

The contrast between these schools is brought out by *Newcomb's Problem*. Causal decision theorists favor the following representation, where two-boxing dominates.

	\$1m in opaque box	Nothing in opaque box
Take both	1,001,000	1,000
Take one	1,000,000	0

Evidential decision theorists can be fine with the following representation, where one-boxing maximises EU so long as one has high enough credence that the predictor got it right:

	Predictor accurate	Predictor inaccurate
Take both	1,000	1,001,000
Take one	1,000,000	0